

Philip Bui

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EDUCATION

Worcester Polytechnic Institute, B.S. in Computer Science, Dean's List, Upsilon Pi Epsilon Aug 2021 - Dec 2023

Tufts University, M.S. in Computer Science: Human-Robot Interaction, Honos Civicus Sep 2024 - May 2025

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer, Cyvl Aug 2025 - Present

- Developing digital twin technologies for cities to improve civil infrastructure and urban planning.

Research Intern, AI Futures Collab, WPI May - Sep 2025

- Investigated the phenomenology and affordances behind human/AI interfaces.

Swarm Robotics AI Research Intern, Air Force Research Laboratory May - Aug 2025

- Developed specialized, heterogeneous multi-agent reinforcement learning algorithms for swarm systems.

Graduate Research Assistant, Human Interaction Lab, Tufts University Jan - May 2025

- Refactored GailBot, a Dockerized Python app for AI transcription and speech analysis with multiple engines.
- Contributed to NextJS deployments via Vercel, integrating TypeScript RESTful APIs with AWS S3 storage.

Full-Stack Software Engineer, enLabel Jan 2024 - Jan 2025

- Architected a React TypeScript frontend with ASP.NET RESTful API integration & end-to-end regression tools.
- Contributed to technical planning and architecture meetings with cross-functional stakeholders & clients.
- Led design specification, timeline management, and strategic planning of engineering deliverables & docs.

Website Developer, ProsperOn Jun 2023 - Feb 2024

- Built React/TypeScript financial simulation web apps & Node.js RESTful APIs, modernizing legacy code.

Operations Intern, Amazon Jun - Aug 2023

- Developed an internal, mapping application, reducing costs by >\$10k (95%) per trained employee.
- Built a Node.js server for data retrieval via Amazon's internal database, located and monitored >7,000 items.

PROJECTS

State-Wise Constrained Policy Shaping: Runtime Behavior Steering for Safe RL, Tufts University Jan - May 2025

- Developed SCPS: a post-training supervisor for RL in HighwayEnv that enforces safety constraints at runtime.
- Reduced collisions by 97% in-distribution and 99% zero-shot; Passed phase 1 of AAI-25: AI Alignment track.

LLM Enhanced Captioning for Neurodivergent Users, Tufts University Jan - May 2025

- Developed a browser extension using Mistral-7B for enriched captions; <200 ms latency & >95% accuracy.

Computer Vision-Guided Sphero Swarm Navigation Framework, Tufts University Sep - Dec 2024

- Developed collision-free PRM path planning & CV MCL SLAM for swarms via BLE & WebSocket microservices.

AI Platform for Low-Income Housing Development for City of Worcester, MA, WPI MQP Aug - Dec 2023

- Designed and implemented AI-driven due diligence analysis, integrating GIS & open source data pipelines.
- Leveraged a PostgreSQL database and Node.js server, using RESTful express APIs for efficient data retrieval.

Nitrogen Cycle Public Outreach AR Game for Boys & Girls Club, WPI IQP Oct 2022 - Mar 2023

- Led an 8-person team to develop an educational React AR.js game with an 86% approval rating from users.

SKILLS

Programming: TypeScript/JavaScript, Python, Java, C, C++, CSS, HTML, C#, SQL, PHP, Swift

Software Development: React, Next.js, Git, RLLib, HuggingFace, OpenCV, NumPy, ASP.NET, PostgreSQL

COMMUNITY EXPERIENCE

Co-Founder & Chief Logistics Officer, Untitled Boston

Dec 2023 - May 2025

Community Events Coordinator, WPI Powerlifting

Jan 2023 - Dec 2023